

STAT 412 QZ07 — Question 3

Problem

Illegal drug use among 12- to 17-year-olds is currently reported as 9.7%. In a random sample of 1,056 students at a large high school, 98 report using illegal drugs. Use $\alpha = 0.05$ to test the principal's claim that illegal drug use in her school is below the current national average.

Solution

The claim is “below,” so use a left-tailed one-proportion test:

$$H_0 : p = 0.097, \quad H_a : p < 0.097.$$

The sample proportion is

$$\hat{p} = \frac{98}{1056} = 0.0928.$$

The test statistic is

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} = \frac{0.0928 - 0.097}{\sqrt{0.097(0.903)/1056}} = -0.46.$$

The left-tailed P-value is

$$P = P(Z \leq -0.46) \approx 0.3225.$$

Since $0.3225 > 0.05$, do not reject H_0 .

Final Answer

Hypotheses: choice B. $z = -0.46$, P-value = 0.3225. Conclusion: choice C; do not reject H_0 . There is insufficient evidence to support the principal's claim.